

2.PHARMACEUTICAL FORMULATION SCIENCE — DEEP DIVE (Part 1)



PHARMACEUTICAL FORMULATION SCIENCE — DEEP DIVE (Part 1)

Dosage Form Mastery + Excipient Science

For: Final Year M.Pharm (Pharmaceutics) → R&D Scientist

Depth: Interview-Ready, Industry-Grade Reference

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PART 1 — DOSAGE FORM MASTERY

1. ORAL SOLID DOSAGE FORMS (OSD)

OSD forms account for **~70% of all pharmaceutical products** and are the #1 focus area for Indian generics companies. This is the bread and butter of formulation R&D.

1.1 TABLETS — Immediate Release (IR)

A) Complete Theory

IR tablets are designed to release the drug rapidly after administration, typically achieving **>85% dissolution within 30 minutes** (USP). The drug disintegrates from the tablet → dissolves in GI fluids → absorbs through GI membrane.

Key Concepts:

- **Compression physics:** Plastic deformation (MCC) vs brittle fracture (lactose, DCP) — bonding mechanisms determine tablet strength
- **Powder flow:** Carr's Index (<15% = excellent), Hausner Ratio (<1.18 = good), Angle of Repose (<30° = excellent)
- **Compactibility:** Tabletability (tensile strength vs compaction pressure), compressibility (porosity vs pressure), compactability (tensile strength vs porosity)
- **Wetting & disintegration:** Capillary action pulls water into tablet → disintegrant swells → tablet breaks → drug dissolves

B) Critical Formulation Parameters

Parameter	Target	Why It Matters
Blend uniformity	RSD ≤5% (USP <905>)	Ensures each tablet has correct dose
Granule flow	Carr's Index <20%	Consistent die filling → consistent weight
Granule moisture	1–3% (typical)	Too wet = sticking; too dry = capping
Compression force	Formulation-dependent	Too high = capping/lamination; too low = friable
Hardness	4–8 kp (typical)	Must survive handling but still disintegrate
Disintegration time	<15 min (USP) for IR	Regulatory requirement
Dissolution	Q ≥80% in 30 min (typical)	Bioavailability assurance
Friability	NMT 1.0% w/w (USP)	Tablet integrity during shipping

C) Key Excipients and Roles

Category	Excipient	Concentration	Role
Diluent/Filler	Microcrystalline Cellulose (MCC, Avicel PH-102)	20–90%	Compressibility, flow (DC grade)
Diluent	Lactose monohydrate (FastFlo 316)	15–75%	Filler, good flow (spray-dried grade for DC)
Diluent	Dibasic calcium phosphate (DCP, Emcompress)	20–50%	Hard, brittle filler

Category	Excipient	Concentration	Role
Binder (dry)	PVP K30 (Povidone)	2–5% (dry), 5–10% in solution	Granule strength
Binder	HPMC E5 (Methocel)	2–5%	Wet granulation binder
Disintegrant	Croscarmellose sodium (Ac-Di-Sol)	2–5% (intragranular + extragranular)	Wicking + swelling mechanism
Disintegrant	Sodium starch glycolate (Explotab)	2–8%	Swelling (200–300x volume)
Disintegrant	Crospovidone (Polyplasdone XL)	2–5%	Wicking + elastic recovery
Lubricant	Magnesium stearate	0.25–1.0%	Reduces die-wall friction; mix only 2–5 min
Glidant	Colloidal silicon dioxide (Aerosil 200)	0.1–0.5%	Improves powder flow
Surfactant	Sodium lauryl sulfate (SLS)	0.5–2.0%	Wetting agent for hydrophobic APIs

D) Manufacturing Process — Direct Compression (DC)

API Sieving (#30–40 mesh)

↓

Excipient Sieving (#30 mesh)

↓

Pre-blending (API + diluent + disintegrant) → V-blender or Bin blender (10–15 min)

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Add glidant (Aerosil) → Blend 3–5 min

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Add lubricant (Mg stearate) → Blend 2–5 min ONLY (over-mixing reduces hardness)

↓

Compression → Rotary tablet press (e.g., Cadmach, Fette)

↓

In-process controls: Weight, Hardness, Thickness, Disintegration

↓

Finished tablet → QC testing

Wet Granulation Process:

API + Diluent + Disintegrant (intragranular) → Dry mix in RMG (2–5 min)

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Add binder solution (PVP in water/IPA) → Granulate in RMG (3–10 min)

↓

Wet screen (#8–12 mesh)

↓

Drying → FBD (Fluid Bed Dryer) at 50–60°C to LOD 1–3%

↓

Dry screen (#20–30 mesh) → Sizing

↓

Add extragranular disintegrant + glidant → Blend 10–15 min

↓

Add lubricant → Blend 2–5 min

↓

E) Quality Control Tests

Test	Method	Acceptance Criteria
Description	Visual	Matches specification (color, shape, embossing)
Identification	HPLC/UV/FTIR	Matches reference standard
Assay	HPLC	90.0–110.0% of label claim (USP typical)
Content Uniformity	USP <905>	AV ≤15.0 (L1 stage); RSD ≤6%
Weight Variation	USP <2091>	±5% for tabs >324mg; ±7.5% for 130–324mg; ±10% for <130mg
Hardness	Tablet hardness tester	Typically 4–8 kp (not official USP, but reported)
Friability	USP <1216>	NMT 1.0% (20 tablets, 100 revolutions, 4 min)
Disintegration	USP <701>	NMT 30 min (uncoated); NMT 15 min (typical target)
Dissolution	USP <711>	Q ≥80% in 30 min (or as per monograph)
Related Substances	HPLC	Per ICH Q3B: individual ≤0.2%, total ≤1.0% (typical)
Moisture	Karl Fischer / LOD	Per specification (typically NMT 2–3%)
Microbial Limits	USP <61>/<62>	TAMC ≤10 ³ CFU/g; TYMC ≤10 ² CFU/g; absent E. coli

F) Common Problems and Troubleshooting

Problem	Root Cause(s)	Solution(s)
Capping	Air entrapment; insufficient binder; excessive fines; low moisture	Increase binder; add pre-compression; reduce speed; optimize LOD
Lamination	Excessive compression force; air entrapment; oily/waxy API	Reduce force; add pre-compression; modify granulation
Sticking/Picking	Excess moisture; low melting point API; punch surface defects	Reduce moisture; add anti-adherent (talc); polish punches; add colloidal SiO ₂

Problem	Root Cause(s)	Solution(s)
Weight Variation	Poor flow; inconsistent die fill; segregation	Improve flow (glidant); optimize particle size; use force-feeder
Low Hardness	Insufficient binder; over-lubrication; excess moisture	Increase binder; reduce Mg stearate mixing time; optimize LOD
High Friability	Insufficient bonding; excess fines	Increase binder; increase compression force; add MCC
Slow Dissolution	Over-lubrication; hydrophobic API; excess hardness	Reduce Mg stearate; add SLS; increase disintegrant; reduce compression
Content Non-uniformity	Segregation; poor mixing; cohesive API	Improve blending; use ordered mixing; granulate the API
Mottling	Dye migration; uneven drug distribution	Use lake dyes (not soluble dyes); improve binder distribution

G) Regulatory Requirements

Aspect	India (CDSCO)	USA (FDA)	EU (EMA)
Pharmacopoeia	IP (Indian Pharmacopoeia)	USP-NF	Ph.Eur. (European Pharmacopoeia)
Filing	DCGI application; CTD format	ANDA (505(j)) or NDA (505(b)(1)/(b)(2)) in eCTD	MAA (Marketing Authorisation Application) in eCTD
Dissolution	IP/USP methods	USP; OGD Dissolution Methods Database	Ph.Eur.; EMA dissolution guidelines
Stability	Zone IVa (30°C/65% RH long-term)	Zone II (25°C/60% RH long-term)	Zone II (25°C/60% RH long-term)
BE Study	Required for generics	Required (in vivo or biowaiver per BCS)	Required (in vivo or biowaiver)
Excipient Status	IP/USP/PhEur listed	Must be in FDA IIG database	Must be in PhEur or justified

H) Top 10 Interview Questions — IR Tablets

Q1: "What factors do you consider when choosing between direct compression and wet granulation?"

Model Answer: I evaluate: (1) API properties — if API has good flow and compressibility (e.g., Metformin HCl at high dose), DC may be possible with DC-grade excipients. If API is cohesive, hygroscopic, or low-dose (<1mg), WG is preferred for content uniformity. (2) Drug-excipient compatibility — moisture-sensitive APIs (e.g., aspirin) favor dry granulation/DC. (3) Dose — high-dose drugs often need WG to achieve acceptable tablet size. (4) Cost — DC has fewer steps, lower cost. (5) Scale-up complexity. In my thesis, I chose WG because my API had poor flow (Carr's Index 32%) and dose was 200mg.

Q2: "Why is magnesium stearate mixed for only 2–5 minutes?"

Over-mixing causes Mg stearate to form a hydrophobic film over granules, reducing interparticle bonding (lower hardness), increasing disintegration time, and decreasing dissolution. Typical mixing is 2–5 min at 12–15 RPM in a V-blender. Over-lubrication is a common cause of dissolution failure at scale-up. I always time lubricant mixing precisely and monitor hardness/dissolution as controls.

Q3: "Your tablet has capping issues at production scale. Walk me through your troubleshooting."

Systematic approach: (1) Check granule moisture (LOD) — if <1%, granules are too dry and brittle → adjust drying endpoint. (2) Check compression parameters — reduce main compression force; increase pre-compression force to deaerate. (3) Check granule PSD — excess fines (<150 µm) cause poor rearrangement → optimize screening/milling. (4) Evaluate binder concentration — increase by 10–20% and run trial. (5) Check punch condition — worn punches cause localized stress. (6) If persistent, consider changing granulation method (e.g., from dry to wet granulation for better bonding). I would run a small-scale DoE varying moisture, pre-compression force, and binder level.

Q4: "Explain the difference between super-disintegrants. When would you choose each?"

Croscarmellose sodium (Ac-Di-Sol): Wicking + swelling. Best all-around; 2–5%. Works well in DC and WG. Effective at low concentrations. **SSG (Explotab):** Extreme swelling (300x). Works well but can affect dissolution if used >8% (gel formation). Best for WG. **Crospovidone (Polyplasdone XL):** Wicking + elastic recovery (returns to original shape). Non-swelling, so doesn't form gel — best for high-dose APIs where gel from SSG might retard dissolution. Choice depends on: (1) API compatibility with moisture (SSG retains water), (2) tablet size constraints, (3) drug release requirements.

Q5: "How do you ensure blend uniformity for a low-dose drug (e.g., 0.5 mg)?"

Challenges: Segregation, poor distribution. Solutions: (1) **Geometric dilution** — premix API with a portion of MCC by doubling. (2) **Ordered mixing** — use fine API particles that adhere

to carrier surface (MCC). (3) **Wet granulation** — dissolve API in binder solution to ensure molecular-level distribution. (4) **Use smaller blender fill** (50–70%) for better mixing. (5) **Validate** using stratified sampling (USP <905> with 10-unit CUSUM). Key: blend uniformity RSD ≤5% at 10 locations (top/middle/bottom, thief sampling).

Q6–Q10 (Brief):

- Q6: "What is the role of pre-compression in a rotary tablet press?" → Deaerates granules, reduces capping/lamination, critical for high-speed presses (>60 RPM)
- Q7: "How does particle size of API affect dissolution and content uniformity?" → Smaller PSD → faster dissolution but poorer flow; needs balance
- Q8: "What is the significance of f1 and f2 values in dissolution?" → f1 (difference factor) ≤15, f2 (similarity factor) ≥50 for profile comparison
- Q9: "Describe USP dissolution Apparatus I vs II" → App I (basket, 100 RPM) for capsules/floating; App II (paddle, 50–75 RPM) for tablets
- Q10: "What is content uniformity vs weight variation? When do you use each?" → CU (USP <905>) for all; WV for >25mg dose and >25% of tablet weight

1.2 TABLETS — Extended Release (ER/SR)

A) Complete Theory

Extended Release (ER) tablets release drug over a prolonged period (typically 8–24 hours) to reduce dosing frequency, minimize side effects, and maintain therapeutic plasma levels. Synonyms: SR (sustained release), CR (controlled release), XR/XL (extended release).

Release Mechanisms:

- **Matrix systems (monolithic):** Drug is dispersed in a polymer matrix; release occurs by diffusion through the matrix and/or erosion of the matrix. Most common in industry.
 - **Hydrophilic matrix:** HPMC (Methocel K100M, K15M) swells to form gel layer → drug diffuses through gel → outer gel erodes. *Example: Glucophage XR (Metformin)*
 - **Hydrophobic matrix:** Ethylcellulose, waxes (Compritol, Precirol). Drug diffuses through insoluble matrix. Erosion-independent. *Example: Certain Theophylline ER*
 - **Lipid matrix:** Glyceryl behenate (Compritol 888 ATO), stearic acid
- **Reservoir systems (membrane-controlled):** Drug core coated with a rate-controlling membrane (ethylcellulose, Eudragit). Release through membrane diffusion.

- Example: Adalat OROS (Nifedipine) — osmotic push-pull

- **Osmotic systems (OROS):** Osmotic pressure pushes drug through laser-drilled orifice. Zero-order release. Example: Procardia XL (Nifedipine), Concerta (Methylphenidate)

Release Kinetics Models:

Model	Equation	When It Applies
Zero-order	$Q = Q_0 + K_0t$	Osmotic systems; ideal ER target
First-order	$\log Q = \log Q_0 - K_1t/2.303$	Conventional release; rate $\propto$ remaining drug
Higuchi	$Q = K_h\sqrt{t}$	Drug release from matrix by diffusion
Korsmeyer-Peppas	$Q = K_{kp}t^n$	Characterizes mechanism: $n=0.45$ (Fickian), $n=0.45-0.89$ (anomalous), $n=0.89$ (Case II/zero-order)
Hixson-Crowell	$Q_0^{1/3} - Qt^{1/3} = K_s t$	Erosion-dominated release

B) Critical Formulation Parameters

Parameter	Target	Why Critical
Polymer type & viscosity grade	HPMC K100M (100,000 cps) for 24h; K15M for 12h	Controls gel strength and erosion rate
Polymer concentration	20–40% w/w for hydrophilic matrix	Below 20% → dose dumping; above 40% → incomplete release
Drug:polymer ratio	Typically 1:0.5 to 1:2	Determines release duration
Particle size of polymer	Fine particle → faster gel; coarse → slower initial release	Affects gel formation speed
Drug solubility	BCS Class I/III preferred	Poorly soluble drugs may need lipid matrix or solubilizer
Tablet geometry	Surface area:volume ratio	Affects release rate
Compression force	Formulation-dependent	Affects porosity → water penetration → release rate

C) Key Excipients for ER

Excipient	Brand	Conc. Range	Role	Marketed Product Example
HPMC K100M	Methocel K100M Premium CR	20–40%	Hydrophilic matrix former (high viscosity)	Glucophage XR (Metformin)
HPMC K15M	Methocel K15M	20–35%	Hydrophilic matrix (medium viscosity)	Generic ER products
HPMC K4M	Methocel K4M	15–30%	Hydrophilic matrix (lower viscosity, shorter release)	—
Ethylcellulose	Ethocel Std 10 Premium	10–30%	Hydrophobic matrix / coating polymer	Theo-Dur (Theophylline)
Eudragit RS/RL	—	5–20% (coating)	Sustained-release coating (pH-independent)	Many pellet/bead systems
Glyceryl behenate	Compritol 888 ATO	10–30%	Lipid matrix former	—
Cetyl alcohol / Stearyl alcohol	—	5–20%	Waxy matrix retardant	—
Xanthan gum	—	10–30%	Natural swelling polymer	—
Polyethylene oxide (PEO)	PolyOx WSR	20–50%	High-swelling matrix; abuse-deterrent formulations	OxyContin (abuse-deterrent)
Cellulose acetate	—	Coating	Osmotic membrane (with PEG as pore former)	OROS systems

D) Manufacturing — Hydrophilic Matrix ER Tablet

API Sieving → #40 mesh
↓
HPMC K100M sieving → #30 mesh (avoid lumps)
↓

Geometric blending: API + HPMC + MCC (diluent) → Bin blender (15–20 min)

↓

Add glidant (Aerosil 200, 0.25%) → Blend 5 min

↓

Add lubricant (Mg stearate 0.5%) → Blend 3 min

↓

Direct Compression OR Wet Granulation → compression on rotary press

↓

In-process: Weight, Hardness (typically 10–15 kp for ER), Thickness

↓

Dissolution testing at multiple timepoints (1, 2, 4, 6, 8, 12h or per profile)

E) QC Tests for ER Tablets

Test	Criteria
Dissolution	Multi-point: e.g., 20–40% at 1h, 40–60% at 4h, ≥80% at 8h (product-specific)
Dissolution Comparison	f2 ≥50 with RLD (Reference Listed Drug)
Assay	90.0–110.0%
Content Uniformity	AV ≤15.0
Related Substances	Per ICH Q3B limits
Water Content	Per specification
Hardness	Typically higher (10–20 kp) for ER

F) Troubleshooting ER Tablets

Problem	Cause	Solution
Dose dumping (burst release)	Insufficient polymer; cracked tablet; alcohol-induced	Increase HPMC%; ensure tablet integrity; test alcohol-induced dose dumping (FDA requirement)
Incomplete release (<80% at terminal timepoint)	Excess polymer; hydrophobic API trapped in gel	Reduce polymer%; add solubilizer (SLS); modify drug:polymer ratio
Variable release profiles	Inconsistent polymer hydration; moisture variation	Control polymer PSD; control granule LOD
Sticking (during compression)	HPMC is sticky when moist	Control humidity; use anti-adherent; avoid excess moisture

1.3 TABLETS — Orally Disintegrating Tablets (ODT)

Key Facts

- Disintegrate on tongue in **<30 seconds** (FDA) or **<3 minutes** (EMA)
- For pediatric, geriatric, psychiatric patients; patients with dysphagia
- **No water needed** for administration
- *Examples: Zyprexa Zydis (Olanzapine), Claritin RediTabs (Loratadine), Zofran ODT (Ondansetron)*

Technologies

Technology	Method	Example Product
Zydis (freeze-drying)	Lyophilized wafer in blister	Zyprexa Zydis
Durasolv	Conventional DC with high superdisintegrant	—
Orasolv	Effervescent + taste-masking	—
FlashTab	Microgranules + superdisintegrant in DC	—
Pharmaburst	Co-processed excipient for DC	Many generics
Direct Compression	High croscopovidone (5–8%) + mannitol (DC grade)	Most generics

Critical Excipients for ODT

Excipient	Role	Concentration
Mannitol (Pearlitol SD 200)	Sugar alcohol filler; pleasant cooling taste	50–80%
Crospovidone (Polyplasdone XL-10)	Fast disintegrant; porous	5–10%
Croscarmellose sodium	Wicking + swelling disintegrant	3–6%
Magnesium stearate	Lubricant	0.5%
Aspartame / Sucralose	Sweetener	0.5–2%
Flavors	Taste masking	0.5–2%
Taste-masking coating (Eudragit E PO)	Polymer coating on drug particles	Depends on API bitterness

Interview Q: "How do you formulate an ODT for a bitter drug?"

Model Answer: Three-pronged approach: (1) **Taste masking** — coat API particles with Eudragit E PO (dissolves at pH <5 in stomach, insoluble at salivary pH 6.8) or ethylcellulose using fluid bed coating. (2) **Cyclodextrin complexation** — HP-β-CD can complex with bitter APIs to block taste receptor interaction. (3) **Ion exchange resin** — Amberlite IRP-69 for drugs like dextromethorphan. Then formulate the taste-masked particles in a DC blend with mannitol (60–70%) + crospovidone (5–8%) targeting <30 sec disintegration. Test using e-tongue or trained human taste panel.

1.4 BILAYER TABLETS

Purpose: Combine two drugs in one tablet, or combine IR + ER layers for a drug, or separate incompatible drugs.

Examples: Glucovance (Metformin + Glyburide), Caduet (Amlodipine + Atorvastatin)

Key Challenges

Challenge	Solution
Layer adhesion (delamination)	Optimize first-layer tamping force; ensure compatible excipients; roughen interface
Cross-contamination	Clean die table between layers; optimize powder flow

Challenge	Solution
Weight control	Requires bilayer tablet press (e.g., Korsch XL 400, Fette 2090i) with separate fill stations
Dissolution interference	Excipients from one layer affecting other layer's release

Interview Q: "What causes delamination in bilayer tablets? How do you fix it?"

Model Answer: Delamination occurs when the bond between layers is weak. Causes: (1) Excessive first-layer compression force (creates a smooth, non-porous surface). Solution: Use low tamping force (1–3 kN) for the first layer — just enough to form a coherent layer. (2) Incompatible excipients creating a weak interface. Solution: Use a common binder in both layers. (3) Excessive lubricant at the interface. Solution: Minimize Mg stearate in both layers. (4) Elastic recovery mismatch between layers. Solution: Reformulate to balance compaction behavior. Testing: Use side-crushing force test to measure inter-layer bonding strength.

1.5 CAPSULES

Hard Gelatin Capsules (HGC)

Parameter	Details
Shell composition	Gelatin (bovine/porcine), water (13–16%), colorants, opacifier (TiO ₂)
Sizes	000 (largest, ~1.37 mL) → 5 (smallest, ~0.13 mL); Size 0 and 1 most common
Fill materials	Powders, pellets, granules, semi-solids, tablets (tablets-in-capsule)
Machine	Automatic capsule filler (Bosch GKF, MG2, IMA Zanasi)
Advantages	Easy to swallow; taste masking; flexible fill combinations
Limitations	Gelatin cross-linking (pellicle formation) with aldehydes; hygroscopic drugs
Example	Omeprazole pellet-filled capsules (enteric-coated pellets in HGC)

HPMC Capsules (Vcaps Plus, DRcaps)

- **Vegetarian** alternative (no animal-derived gelatin)

- **Lower moisture content** (4–6% vs 13–16% for gelatin) → better for hygroscopic/moisture-sensitive APIs
- **DRcaps:** Delayed-release HPMC capsules (resist acid, open in intestine) — no need for enteric coating of pellets
- *Growing rapidly in India and globally*

Soft Gelatin Capsules (SGC)

Parameter	Details
Shell	Gelatin + plasticizer (glycerin/sorbitol) + water
Fill	Liquids, pastes, suspensions in oils or PEG
Key use	Lipophilic drugs, SEDDS/SNEDDS, vitamins (A, D, E, K)
Manufacturing	Rotary die process (seamless or die-formed)
Example	Neoral (Cyclosporine SNEDDS in SGC), Accutane (Isotretinoin)
Challenge	Drug migration into shell; shell-fill compatibility

1.6 PELLETS & MULTIPARTICULATES

Theory

- **Pellets** = small spherical particles (0.5–2.0 mm) containing drug, prepared by extrusion-spheronization or layering
- **Multiparticulates** = multiple-unit dosage form → each pellet is an independent delivery unit
- **Advantage over monolithic:** More uniform GI transit; less risk of dose dumping; flexible release profiles
- *Example: Omeprazole DR capsules (enteric-coated pellets), Toprol-XL (pellets in tablet)*

Manufacturing Methods

Method	Process	Use
Extrusion-Spheronization	Wet mass → extruder (screen 0.7–1.0mm) → spheronizer (2–5 min, 700–1200 RPM) → drying (FBD)	Drug-loaded pellets
Drug Layering	Sugar/MCC spheres (Nonpareils) → coat with drug + binder solution in Wurster coater	Low-dose, high-potency drugs

Method	Process	Use
Wurster Coating	Bottom-spray fluid bed coating → apply functional coating (enteric, SR)	Functional film coating

Coating Types for Pellets

Coating	Polymer	Purpose	Example
Enteric	Eudragit L30D-55 (dissolves at pH >5.5), HPMC phthalate	Acid protection	Omeprazole pellets
Sustained release	Eudragit RS/RL 30D (pH-independent), Ethylcellulose (Surelease)	Extended release	Diltiazem ER capsules
Taste masking	Eudragit E PO	Prevents bitter taste in mouth	Clarithromycin granules

2. LIQUID DOSAGE FORMS

2.1 SOLUTIONS

Oral Solutions: Drug completely dissolved. Clear. Example: *Paracetamol Syrup IP*

Key Formulation Considerations

Parameter	Consideration
Solubility	Drug must be fully dissolved at all temperatures (including cold chain)
pH	Buffer for stability and taste; check compatibility
Preservatives	Required for multi-dose: methylparaben (0.02%) + propylparaben (0.01%) or sodium benzoate (0.1–0.2%)
Sweeteners	Sucrose (50–65% w/v), sorbitol (25–35%), sucralose (0.01–0.05%)
Viscosity	CMC-Na, xanthan gum for mouthfeel
Stability	Light, oxidation, hydrolysis → use antioxidants, amber bottles

2.2 SUSPENSIONS

Definition: Coarse dispersion of insoluble drug particles (1–50 µm) in a liquid vehicle. *Example:* Amoxicillin oral suspension

Key Concepts

- **Deflocculated:** Particles settle slowly as a hard cake (difficult to redisperse) → NOT desirable
- **Flocculated:** Particles form loose aggregates (flocs) → settle fast but redisperse easily → DESIRABLE
- **Structured vehicle:** Suspending agent creates a network that holds particles → prevents settling

Critical Excipients

Excipient	Role	Concentration
Xanthan gum	Suspending/thickening (pseudoplastic)	0.1–0.5%
CMC-Na (Sodium CMC)	Suspending agent	0.5–2.0%
Veegum (Magnesium aluminum silicate)	Suspending, thixotropic	0.5–2.0%
Tween 80 (Polysorbate 80)	Wetting agent	0.1–0.5%
Methylparaben + Propylparaben	Preservative system	0.18% + 0.02%
Citric acid + Sodium citrate	Buffer	pH 4–6 typical

Interview Q: "Why do we prefer flocculated suspensions?"

Flocculated suspensions have loosely bonded particle aggregates that settle rapidly but form a soft, easily redispersible sediment. Deflocculated suspensions settle slowly but form a hard, compact cake that is nearly impossible to redisperse — this is dangerous because the patient gets a variable dose. We control flocculation using electrolytes (NaCl), surfactants, or oppositely charged polymers to create controlled, reversible aggregation.

2.3 EMULSIONS

Type	System	Example	Surfactant HLB
O/W	Oil droplets in water	Most oral/topical products	HLB 8–18

Type	System	Example	Surfactant HLB
W/O	Water droplets in oil	Some topical creams (cold cream)	HLB 3–6
Microemulsion	Thermodynamically stable, transparent	Neoral (Cyclosporine)	Requires surfactant + co-surfactant
Nanoemulsion	Kinetically stable, droplet <200nm	Parenteral lipid emulsions	—

3. PARENTERAL DOSAGE FORMS

3.1 Small Volume Parenterals (SVPs)

Critical Requirements

Parameter	Requirement	Why
Sterility	Must be STERILE (no viable microorganisms)	Direct injection bypasses all body defenses
Pyrogen-free	Endotoxin limits: ≤ 5 EU/kg (IV); ≤ 0.2 EU/mL (intrathecal)	Endotoxins cause fever, shock
Particulate matter	USP <788>: ≤ 6000 particles $\geq 10\mu\text{m}$ /container; ≤ 600 particles $\geq 25\mu\text{m}$ /container	Embolism, phlebitis
Isotonicity	~ 290 mOsm/kg (0.9% NaCl equivalent)	Prevent hemolysis or crenation
pH	3.0–10.5 tolerable; target 4.0–8.0	Tissue irritation
Clarity	Solution must be clear, particle-free	Visual inspection required

Key Excipients for Parenterals

Excipient	Function	Typical Use
Water for Injection (WFI)	Vehicle	Must meet USP WFI specs (endotoxin <0.25 EU/mL)
NaCl (0.9%)	Tonicity adjuster	Isotonic solution
Dextrose (5%)	Tonicity + caloric value	D5W

Excipient	Function	Typical Use
Polysorbate 80 (Tween 80)	Surfactant/solubilizer	0.01–0.1% for biologics
Sodium hydroxide / HCl	pH adjustment	To target pH
Buffers (phosphate, citrate, acetate)	pH stability	Product-specific
EDTA (Edetate disodium)	Chelating agent (antioxidant synergist)	0.01–0.075%
Sodium metabisulfite	Antioxidant	0.01–0.1% (note: sulfite sensitivity)
Benzyl alcohol	Preservative (multi-dose)	0.9–2.0% (Contraindicated in neonates)
Mannitol / Sucrose / Trehalose	Lyoprotectant / bulking agent	2–5% for lyo products

Sterilization Methods

Method	Condition	When Used
Terminal sterilization (autoclave)	121°C, 15 psi, 15–20 min	Heat-stable products (first choice per FDA/EMA)
Sterile filtration	0.22 µm membrane filter	Heat-labile solutions (most proteins, peptides)
Aseptic processing	Sterile manufacturing environment (Class 100/ISO 5)	When terminal sterilization and filtration are not feasible
Radiation	Gamma / e-beam	Some devices, raw materials

3.2 Lyophilized Products

Freeze-Drying Cycle Design

Step 1: FREEZING (–40 to –50°C, 2–4 hours)
→ Product freezes; ice crystals form; solutes concentrate between ice crystals
→ Must freeze below Tg' (glass transition of maximally freeze-concentrated solute)

→ Annealing step (optional): hold at -10 to -20°C to grow larger ice crystals (faster sublimation)

↓

Step 2: PRIMARY DRYING (Shelf temp: -20 to -10°C; Chamber pressure: 50–200 mTorr)

- Ice sublimates (solid → vapor) under vacuum
- Longest step (12–48+ hours)
- Product temp must stay below T_c (collapse temperature)
- Monitor by Pirani gauge, thermocouple, or comparative pressure measurement

↓

Step 3: SECONDARY DRYING (Shelf temp: +25 to +40°C; same vacuum)

- Removes bound/unfrozen water from amorphous matrix
- Target: residual moisture <1–3% for stability
- Duration: 3–8 hours

↓

Step 4: STOPPERING (under vacuum or nitrogen backfill)

- Vials stoppered inside lyophilizer
- Crimped with aluminum seals

Key Lyo Excipients

Excipient	Role	Typical Conc.
Sucrose	Lyoprotectant (stabilizes protein by water replacement)	2–10%
Trehalose	Lyoprotectant (higher T _g than sucrose)	2–10%

Excipient	Role	Typical Conc.
Mannitol	Bulking agent (forms crystalline cake)	2–5%
Glycine	Bulking agent + tonicity	1–3%
Polysorbate 80	Surfactant (prevents protein aggregation)	0.01–0.05%
Histidine / Phosphate buffer	pH control	10–50 mM

Interview Q: "Why do we use sucrose or trehalose in lyophilized biologics?"

Model Answer: During freeze-drying, removal of the hydration shell from proteins can cause unfolding and aggregation. Sucrose and trehalose act as **lyoprotectants** via two mechanisms: (1) **Water replacement hypothesis** — sugar hydroxyl groups form hydrogen bonds with protein surface, replacing water molecules and maintaining native conformation. (2) **Vitrification** — they form an amorphous glass matrix (high Tg) that immobilizes protein molecules, preventing conformational changes during storage. Trehalose is often preferred over sucrose because it has a higher Tg (115°C vs 62°C for sucrose), allowing for higher secondary drying temperatures and better long-term stability. Key: the sugar:protein molar ratio should typically be >300:1 for adequate stabilization.

4. SEMI-SOLID DOSAGE FORMS

4.1 Ointments

Classification of Ointment Bases

Base Type	Examples	Properties	Use
Hydrocarbon (oleaginous)	White petrolatum, mineral oil	Occlusive; water-repellent; emollient	Dry skin; protective barrier
Absorption bases	Hydrophilic petrolatum, lanolin	Can absorb water (w/o emulsifier present)	Incorporating aqueous solutions into oily base
Water-removable (o/w creams)	Hydrophilic ointment USP	Washable; non-greasy	Most drug delivery applications
Water-soluble	PEG ointment (PEG 400 + PEG	No oil/water; washable	Wounds; water-soluble drugs

Base Type	Examples	Properties	Use
	3350)		

4.2 Creams and Gels

Gels Classification

Type	Base	Example
Hydrogels	Carbomer (Carbopol 940, 0.5–2%), HPMC, CMC-Na	Voltaren Gel (diclofenac)
Organogels	Colloidal MgAl silicate in propylene glycol	Some topical preparations

Transdermal Patches

Type	Design	Example Product
Matrix	Drug dispersed in polymer matrix (PSA + drug)	Nicotine patch (Nicoderm CQ)
Reservoir	Drug solution in a compartment; rate-controlling membrane	Estradiol patch (Estraderm)
Drug-in-Adhesive (DIA)	Drug dissolved/dispersed in PSA layer	Fentanyl patch (Duragesic), Lidocaine (Lidoderm)

Key Parameters

- **Flux (J)** = Amount of drug permeating per unit area per time ($\mu\text{g}/\text{cm}^2/\text{h}$)
- **Permeation enhancers:** Oleic acid, propylene glycol, azone, terpenes
- **Adhesive:** Pressure-sensitive adhesive (PSA) — acrylate, silicone, PIB (polyisobutylene)
- **Backing layer:** Polyester, polyethylene — occlusive or breathable
- **Release liner:** Silicone-coated polyester — removed before application

5. ADVANCED DRUG DELIVERY SYSTEMS (ADDS)

5.1 Nanoparticles

Polymeric Nanoparticles

- **Size:** 10–1000 nm (typically 100–300 nm for drug delivery)
- **Polymers:** PLGA, PLA, PCL, chitosan, albumin
- **Methods:** Nanoprecipitation, emulsion-solvent evaporation, double emulsion (w/o/w)
- **Example: Abraxane** (nab-paclitaxel) — albumin-bound paclitaxel nanoparticles (~130 nm), eliminates need for Cremophor EL solvent

Solid Lipid Nanoparticles (SLN)

- Solid lipid core (e.g., glyceryl monostearate, Compritol) stabilized by surfactant
- Size: 50–1000 nm
- Advantage: Biocompatible lipids; controlled release; can be lyophilized

5.2 Liposomes

Type	Composition	Size	Example
Conventional	Phosphatidylcholine + cholesterol	100–400 nm	Ambisome (Amphotericin B)
Stealth (PEGylated)	+ DSPE-PEG 2000 (5–10 mol%)	80–120 nm	Doxil (PEGylated liposomal doxorubicin)
Cationic	+ DOTAP or DOTMA	50–200 nm	Gene delivery
Thermosensitive	+ DPPC (transition at ~41°C)	80–150 nm	ThermoDox (in clinical trials)

Interview Q: "What is the difference between liposomes and LNPs?"

Model Answer: Both are lipid-based nanovesicles but differ fundamentally. **Liposomes** have a well-defined **lipid bilayer** enclosing an aqueous core — hydrophilic drugs go in the core, lipophilic drugs in the bilayer. **LNPs** (Lipid Nanoparticles) have a more **complex internal structure** — they are NOT simple bilayer vesicles. LNPs typically contain ionizable cationic lipids that form an inverted hexagonal phase encapsulating nucleic acid (mRNA/siRNA) in the core. Key compositional difference: LNPs use **ionizable lipids** (e.g., DLin-MC3-DMA, ALC-0315, SM-102) that are neutral at physiological pH (avoiding toxicity) but become positively charged at endosomal pH (~5) to facilitate endosomal escape. LNPs are the delivery platform for COVID-19 mRNA vaccines (Comirnaty, Spikevax).

5.3 Lipid Nanoparticles (LNPs)

Composition (Four-Component System)

Component	Function	Example	Typical Mol%
Ionizable cationic lipid	Encapsulates nucleic acid; enables endosomal escape	SM-102 (Moderna), ALC-0315 (Pfizer/BioNTech)	30–50%
Helper phospholipid	Structural stability of bilayer	DSPC (distearoylphosphatidylcholine)	10–20%
Cholesterol	Membrane rigidity and stability	Cholesterol	25–40%
PEG-lipid	Steric stabilization; controls particle size; extends circulation	ALC-0159, DMG-PEG 2000	1–5%

Manufacturing: Microfluidic Mixing

Ethanol phase (lipids dissolved in ethanol)

↓

→ T-junction or staggered herringbone mixer ←

↓

↓

Aqueous phase (mRNA in citrate buffer pH 4.0)

↓

Rapid mixing → LNP self-assembly (~50–100 nm)

↓

Dialysis/TFF → Remove ethanol, raise pH to 7.4

↓

Sterile filtration (0.22 µm)

↓

Fill into vials → Store at -20°C or -70°C (mRNA stability)

Key Characterization

Parameter	Method	Target
Particle size	DLS (Dynamic Light Scattering)	60–100 nm; PDI <0.2
Encapsulation efficiency	RiboGreen assay	>85%
Zeta potential	Electrophoretic mobility	Near neutral at pH 7.4
mRNA integrity	Capillary electrophoresis	>70% intact
Endotoxin	LAL assay	<5 EU/kg

5.4 PLGA-Based Microspheres

PLGA (Poly(lactic-co-glycolic acid)) — FDA-approved biodegradable polymer

Marketed Examples

Product	Drug	Release Duration	Indication
Lupron Depot	Leuprolide acetate	1, 3, 4, or 6 months	Prostate cancer, endometriosis
Risperdal Consta	Risperidone	2 weeks	Schizophrenia
Vivitrol	Naltrexone	1 month	Alcohol/opioid dependence
Bydureon	Exenatide	1 week	Type 2 diabetes

PLGA Degradation

- **Mechanism:** Bulk hydrolysis of ester bonds → lactic acid + glycolic acid (Krebs cycle metabolites) → $\text{CO}_2 + \text{H}_2\text{O}$
- **Factors affecting degradation:** Lactide:glycolide ratio (50:50 degrades fastest; 85:15 slowest), molecular weight, end-cap (acid-terminated degrades faster than ester-terminated)
- **Drug release:** Triphasic — initial burst (surface drug) → lag phase (slow diffusion through intact polymer) → erosion phase (rapid release as polymer disintegrates)

5.5 Hydrogels

- **Definition:** 3D cross-linked polymer networks that swell in water but do not dissolve
- **Types:** Chemical (covalent cross-links, permanent) vs Physical (ionic, H-bonding, reversible)

- **Stimuli-responsive:** pH-sensitive (Eudragit), temperature-sensitive (PNIPAAm, LCST ~32°C), enzyme-responsive
- **Applications:** Wound dressings, ophthalmic inserts, injectable depots, tissue engineering

PART 2 — EXCIPIENT SCIENCE: TOP 50 EXCIPIENTS

BINDERS

#	Excipient (Brand)	Mechanism	Conc. Range	Key Compatibility Issues	Regulatory	Interview Tip
1	PVP K30 (Povidone) (Kollidon 30)	H-bonding between particles	2–5% (dry); 5–15% (solution)	Hygroscopic; may reduce dissolution if excess	GRAS, IIG, PhEur	"Why PVP over HPM as binder — PVP gives stronger granules HPMC g more ele
2	HPMC E5/E15 (Methocel E5)	Film-forming + adhesion	2–5% (wet binder)	Less hygroscopic than PVP; slower dissolution of binder film	GRAS, IIG, PhEur	"Which grade fo binding v matrix?" E-grade binding, grade fo SR matr
3	PVP-VA (Copovidone) (Kollidon VA 64)	H-bonding; less hygroscopic than PVP	2–5% (dry); used in HME	Key ASD polymer; Tg ~101°C	IIG, PhEur	"Why for HME?" - Lower Tg than PVI K30 (149°C),

#	Excipient (Brand)	Mechanism	Conc. Range	Key Compatibility Issues	Regulatory	Interview Tip
						process at lower temp
4	Pregelatinized starch (Starch 1500)	Swelling + adhesion	5–20%	Dual function: binder + disintegrant at different concentrations	GRAS, IIG	"How do differ from regular starch?" Cold-water soluble, directly compressible
5	HPC (Hydroxypropyl cellulose) (Klucel)	Film-forming	2–6% (binder); 15–35% (coating)	Thermoplastic; used in HME	IIG, PhEur	Less common than HPMC/F as binder

DILUENTS / FILLERS

#	Excipient (Brand)	Mechanism	Conc. Range	Key Issues	Regulatory	Interview Tip
6	MCC (Microcrystalline Cellulose) (Avicel PH-102, PH-101)	Plastic deformation → strong bonds; moisture buffer	20–90%	Moisture absorption; PH-101 for WG, PH-102 for DC	GRAS, IIG, PhEur	#1 most common excipient "What is PH-102 vs PH-101?" — PH-102 large particle size (100µm) for flow
7	Lactose Monohydrate (FastFlo 316, Tablettose)	Brittle fracture	15–75%	Incompatible with amines (Maillard reaction → browning); contraindicated in lactose intolerance	GRAS, IIG, PhEur	"When NOT use lactose?" — Amines (aminoplastics) lactose-intolerant patients

#	Excipient (Brand)	Mechanism	Conc. Range	Key Issues	Regulatory	Interview
8	Mannitol (Pearlitol SD 200)	Non-hygroscopic; pleasant taste	10–90%	Negative heat of solution (cooling sensation); more expensive than lactose	GRAS, IIG, PhEur	Preferred ODT; "Why mannitol ODTs?" – Sweet, non-hygroscopic
9	DCP (Dibasic Calcium Phosphate) (Emcompress)	Hard, brittle filler	20–50%	Abrasive to tooling; alkaline pH → incompatible with acid-sensitive APIs	GRAS, IIG, PhEur	"Advantages DCP?" – Excellent high density, low moisture
10	Starch (Maize/Corn Starch)	Filler + disintegrant at high%	10–75%	Poor compressibility alone; hygroscopic	GRAS, IIG	Traditional use, modern formulations

DISINTEGRANTS

#	Excipient (Brand)	Mechanism	Conc. Range	Key Issues	Regulatory	Interview Tip
11	Croscarmellose Sodium (Ac-Di-Sol)	Wicking + swelling	2–5% (intra + extra)	Sodium content (for sodium-restricted patients)	IIG, PhEur	"Best all-around super-disintegrant" — effective at low conc.
12	SSG (Sodium Starch Glycolate) (Explotab, Primojel)	Massive swelling (200–300x)	2–8%	>8% → forms gel → retards dissolution	IIG, PhEur	"Why limit SSG to <8%?" → Gel formation
13	Crospovidone (Polyplasdone XL, XL-10)	Wicking + elastic recovery (non-swelling)	2–5%	Porous; good for ODT (fast wicking)	IIG, PhEur	"When to choose crospovidone over CCS?" — When gel formation from

#	Excipient (Brand)	Mechanism	Conc. Range	Key Issues	Regulatory	Interview Tip
						SSG/CCS would retard dissolution of high-dose drug
14	Low-substituted HPC (L-HPC) (LH-11, LH-21)	Swelling	5–15%	Also acts as binder; popular in Japan	IIG, PhEur	Less common in India/USA, very common in Japanese formulations

LUBRICANTS & GLIDANTS

#	Excipient (Brand)	Mechanism	Conc. Range	Key Issues	Regulatory	Interview Tip
15	Magnesium Stearate	Boundary lubricant (metal stearate film on particles)	0.25–1.0%	#1 problem excipient: Hydrophobic → reduces hardness, dissolution; over-mixing is fatal; batch-to-batch variability in specific surface area	GRAS, IIG, PhEur	Most asked lubricant question: "Why mix only 2-5 min?"
16	Stearic Acid	Lubricant	1–3%	Less effective than Mg stearate; less hydrophobic	GRAS, IIG	Alternative when Mg stearate is incompatible
17	Sodium Stearyl Fumarate (PRUV)	Lubricant	0.5–2.0%	Water-soluble → less impact on dissolution; more expensive	IIG, PhEur	"What lubricant has least impact on dissolution?" → SSF/PRUV

#	Excipient (Brand)	Mechanism	Conc. Range	Key Issues	Regulatory	Interview Tip
18	Colloidal Silicon Dioxide (Aerosil 200, Cab-O-Sil)	Glidant (reduces interparticle friction)	0.1–0.5%	Fine powder; absorbs moisture	GRAS, IIG, PhEur	"What is the difference between glidant and lubricant?" — Glidant improves FLOW; lubricant reduces FRICTION with die wall
19	Talc	Glidant + anti-adherent	1–5%	Potential asbestos contamination concern; FDA scrutiny	GRAS, IIG	Declining in use due to safety concerns; check USP limits

COATING AGENTS

#	Excipient (Brand)	Function	Conc. Range	Key Issues	Regulatory
20	HPMC-based film coat (Opadry, Opadry II)	Immediate-release film coat (aesthetic, moistureprotection)	2–4% weight gain	Ready-to-use premix; just add water	IIG, PhEur
21	Eudragit L100-55 / L30D-55	Enteric coat (dissolves at pH >5.5)	4–8 mg/cm ²	Requires plasticizer (TEC, triethyl citrate)	IIG, PhEur
22	Eudragit S100	Enteric (dissolves at pH >7.0 → colon targeting)	4–8 mg/cm ²	Slower dissolution than L grades	IIG, PhEur
23	Ethylcellulose (Surelease, Aquacoat ECD)	Sustained-release coat (water-insoluble)	5–15% weight gain	Requires pore former (HPMC/PEG); needs curing	IIG, PhEur
24	Eudragit RS/RL PO or 30D	Sustained-release coat (pH-	5–15% weight	RS:RL ratio controls	IIG, PhEur

#	Excipient (Brand)	Function	Conc. Range	Key Issues	Regulatory
		independent)	gain	permeability (RS=less permeable)	
25	PVA-based (Opadry AMB)	Moisture barrier coat	2–4% weight gain	Lower moisture permeability than HPMC	IIG

SUSTAINED-RELEASE POLYMERS

#	Excipient (Brand)	Mechanism	Conc. Range	Key Drug Example	Regulatory
26	HPMC K100M CR (Methocel K100M)	Hydrophilic matrix → gel barrier	20–40%	Glucophage XR (Metformin ER)	IIG, PhEur
27	HPMC K15M CR (Methocel K15M)	Hydrophilic matrix (medium viscosity)	20–35%	Many generic ER products	IIG, PhEur
28	Polyethylene Oxide (PEO) (PolyOx WSR Coagulant, N-60K)	Swelling + erosion matrix	20–50%	OxyContin (abuse-deterrent ER)	IIG
29	Ethylcellulose (Ethocel Std 10 or 20)	Insoluble matrix / coating	10–30% (matrix)	Theo-Dur (Theophylline)	IIG, PhEur
30	Carbomer (Carbopol 971P)	Hydrophilic swelling polymer	5–30%	Some ER tablets	IIG, PhEur

SOLUBILIZERS & SURFACTANTS

#	Excipient (Brand)	Function	Conc. Range	Key Notes	Regulatory
31	SLS (Sodium Lauryl Sulfate)	Anionic surfactant; wetting	0.5–2.0%	Can affect dissolution testing media; may interact with cationic drugs	IIG, PhEur

#	Excipient (Brand)	Function	Conc. Range	Key Notes	Regulatory
32	Polysorbate 80 (Tween 80)	Non-ionic surfactant; solubilizer	0.1–5% (oral); 0.01–0.1% (parenteral)	Auto-oxidation → peroxides → protein degradation	IIG, PhEur
33	Poloxamer 188 (Pluronic F-68)	Non-ionic block copolymer surfactant	0.1–5%	Used in nasal sprays, parenteral, oral	IIG, PhEur
34	Cremophor EL (Polyoxyl 35 castor oil)	Solubilizer	5–50%	Hypersensitivity reactions (anaphylaxis); being replaced by alternatives	IIG
35	HP-β-Cyclodextrin (Cavitron, Kleptose HPB)	Inclusion complexation	1:1 to 1:2 molar ratio	Can affect drug absorption; renal toxicity of parent β-CD (not HP-β-CD)	IIG, PhEur
36	SBE-β-Cyclodextrin (Captisol)	Inclusion complexation (parenteral-safe)	Up to 40% w/v	Used in Vfend IV (voriconazole)	IIG
37	Soluplus (PVA-PEG graft copolymer)	Polymeric solubilizer; ASD matrix	10–30%	Excellent for HME (T _g ~70°C)	IIG

PARENTERAL-SPECIFIC EXCIPIENTS

#	Excipient	Function	Conc. Range	Key Notes	Regulatory
38	Mannitol	Bulking agent (lyo); tonicity adjuster	2–5% (lyo); 5% (tonicity)	Crystallizes during lyo → elegant cake	IIG, PhEur
39	Sucrose	Lyoprotectant	2–10%	T _g ' = -32°C; amorphous in lyo cake	IIG, PhEur
40	Trehalose	Lyoprotectant (higher T _g than sucrose)	2–10%	T _g = 115°C (dry); preferred for protein stability	IIG, PhEur

#	Excipient	Function	Conc. Range	Key Notes	Regulatory
41	Histidine buffer	pH buffering (pH 5.5–6.5)	10–25 mM	Preferred for mAb formulations	IIG, PhEur
42	Arginine	Aggregation inhibitor; viscosity reducer	50–200 mM	Used in high-concentration mAb formulations	IIG
43	Benzyl alcohol	Preservative (multi-dose vials)	0.9–2.0%	Contraindicated in neonates (gasping syndrome)	IIG, PhEur

PRESERVATIVES & ANTIOXIDANTS

#	Excipient	Function	Conc. Range	Key Notes	Regulatory
44	Methylparaben	Antimicrobial preservative	0.015–0.2%	Often combined with propylparaben (paraben system)	IIG, PhEur
45	Propylparaben	Antimicrobial preservative	0.01–0.02%	Synergistic with methylparaben	IIG, PhEur
46	BHT (Butylated Hydroxytoluene)	Lipid antioxidant	0.005–0.02%	For oils, lipid formulations	GRAS, IIG
47	Ascorbic acid	Aqueous antioxidant	0.01–0.5%	pH-dependent effectiveness; may cause browning	GRAS, IIG

SWEETENERS & FLAVORS

#	Excipient	Function	Conc. Range	Key Notes	Regulatory
48	Sucralose	Artificial sweetener (600x sucrose)	0.01–0.05%	Heat-stable; non-cariogenic	GRAS, IIG
49	Aspartame	Artificial sweetener	0.5–2.0%	Contraindicated in PKU; unstable at high pH/temp	GRAS, IIG

#	Excipient	Function	Conc. Range	Key Notes	Regulatory
50	Peppermint / Strawberry flavor	Taste masking	0.1–1.0%	Must be pharma-grade; natural or artificial	IIG

Important

Interview Master Tip: When asked "Which excipient would you use for X?", always give:

1. The specific excipient name and brand
2. The concentration range
3. WHY you chose it (mechanism)
4. What compatibility issue to check
5. What alternative you'd consider if #1 doesn't work

Example: "For a disintegrant, I'd use **croscarmellose sodium (Ac-Di-Sol) at 3% intragranular + 2% extragranular**. It works by wicking and swelling. I'd verify compatibility with the API via DSC/accelerated stability. If CCS doesn't give adequate disintegration (<15 min), I'd switch to **crospovidone (Polyp lasdone XL) at 5%** which uses a non-swelling wicking/elastic recovery mechanism."

End of Part 1. Continue to Part 2 for Formulation Development Process, Solubility Enhancement, and Practical Lab Tasks.